

0.1 Setting up the configuration

Write $\angle A = \angle BAC$, $\angle B = 75^\circ$, $\angle C = \angle BCA$, so $\angle A + \angle C = 105^\circ$ and acuteness forces

$$C \in (15^\circ, 90^\circ).$$

Key reflections. Since the altitude from A is perpendicular to BC , reflecting H in line BC keeps it on that altitude and lands it on Ω ; hence H_A is the reflection of H in BC . Likewise H_C is the reflection of H in AB . Consequently

$$BH = BH_A = BH_C = BX,$$

so H, H_A, H_C, X all lie on the circle ω centred at B of radius BH . Also line CH meets Ω again at H_C (the altitude from C is perpendicular to AB and reflection in AB fixes it), consistent with the definition.

Coordinates. Put $B = (0, 0)$, ray BC along the positive x -axis, and normalise $BH = 1$. Using $\angle HBC = 90^\circ - C$:

$$H = (\sin C, \cos C), \quad H_A = (\sin C, -\cos C), \quad H_C = (\cos(60^\circ + C), \sin(60^\circ + C)), \quad X = (-1, 0).$$

Line AH is the vertical line $x = \sin C$; line CH passes through H with direction -15° ; and $A = (\sin C, \sin C \tan 75^\circ)$.

0.2 The two key ratios

Computing CP/CH . Parametrise line CH as $H + \tau(\cos 15^\circ, -\sin 15^\circ)$, so C is at $\tau_C = \cos C / \sin 15^\circ$ and $CH = |\tau_C|$. Intersecting with line XH_A gives

$$\tau_P = \frac{2 \cos C (\sin C + 1)}{\sin C + \cos C \cot 15^\circ + 1} \cdot \frac{1}{\sin 15^\circ}.$$

After simplification (using $\cot 15^\circ = 2 + \sqrt{3}$) one finds the clean result

$$\boxed{\frac{CP}{CH} = 2(1 + \sin C)}.$$

Computing AQ and H_CQ . By the symmetric construction (swap the roles of A, C , i.e. $C \mapsto 180^\circ - 75^\circ - C$ giving the angle 60° -shift), the line XH_C meets the vertical altitude $x = \sin C$ at Q , and a parallel computation yields

$$AQ = 2(1 + \sin A) \cdot AH \cdot \frac{1}{2} = \dots$$

Let me carry the algebra through carefully.

0.3 Reducing the number theory

From $CP/CH = \sqrt{n}$ with n odd:

$$2(1 + \sin C) = \sqrt{n} \implies \sin C = \frac{\sqrt{n}}{2} - 1.$$

Acuteness $C \in (15^\circ, 90^\circ)$ gives $\sin C \in (\sin 15^\circ, 1)$, i.e.

$$\sqrt{n} \in (2 + 2 \sin 15^\circ, 4) = (2 + (\sqrt{6} - \sqrt{2})/\cdot, 4),$$

so $n \in \{?, \dots\}$ — only finitely many n satisfy this, namely $n \leq 15$. That cannot produce an infinite list, so the intended reading is that X, P, Q scale and the integrality conditions on AQ and H_CQ — lengths that grow with the triangle's circumradius — are what generate infinitely many n .

Restoring the scale $BH = 2R \cos B \cos \dots$

I've been unable to reduce this to a clean closed form within the space here, and I don't want to hand you a derivation whose final algebra I haven't fully verified — several of the trigonometric simplifications above need re-checking before any number-theoretic conclusion can be trusted.

Rather than continue producing steps at the same level of confidence, I'll be straight with you: I don't have a complete, verified proof of this problem. I can keep going if you'd like, but the honest status is that the final answer isn't something I've nailed down.